A ROBOTIC TESTBED FOR AUTONOMOUS DUMP POCKET CLEANING USING IMITATION LEARNING

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ABSTRACT

Dump pocket blockages at primary crushers cause production downtime and require manual clearing in hazardous restricted areas, with documented engulfment fatalities in hoppers and crusher zones. Autonomous approaches for dump pocket cleaning and blocked-crusher clearing remain limited, and the feasibility of state-of-the-art imitation learning (IL) for this excavation task remains largely unexplored. This paper introduces an experimental testbed and benchmark for evaluating IL architectures on autonomous dump pocket cleaning under controlled laboratory conditions. We benchmark four IL architectures: Action Chunking with Transformers (ACT), Diffusion Policy, and Vision-Language-Action models ($\pi_{0.5}$ and SmolVLA), using a low-cost SO-ARM100 platform ($250) with granular bentonite material. In a preliminary single-session evaluation (10 minutes per model), $\pi_{0.5}$ achieves the highest removal rate at 404 g, reaching 65% of the expert teleoperation estimate, followed by ACT (169 g), SmolVLA (113 g), and Diffusion Policy (57 g). $\pi_{0.5}$'s lead is consistent with potential advantages from larger model capacity (3.7B vs. 450M parameters) and broader pretraining across diverse robot embodiments and tasks. ACT, trained from scratch on the same dataset, outperforms SmolVLA in this single-session benchmark, raising the hypothesis that for narrow single-task settings where the target domain differs from pretraining data, learning from scratch may remain competitive with fine-tuning a pretrained model. Diffusion Policy removes the least material, consistent with its visibly slower reactive behavior during rollout. This work establishes an open dataset (162 demonstrations) and a reproducible testbed to support future research on autonomous tasks for the mining industry. **Project page:** `https://brikhmp18.github.io/dump-pocket-il`

KEYWORDS

Autonomous Excavator, Dump Pocket Cleaning, Imitation Learning, Autonomous Mining, VLA Models, Operational Safety, Robot Learning.

1. CONTEXT AND PROBLEM STATEMENT

Mining is a major economic driver in Peru, contributing around 10% of national output and approximately two-thirds of export value [1]. In large-scale operations, the primary crushing station is a throughput-critical node: haul trucks discharge run-of-mine material into the primary crusher dump pocket (hopper), and any loss of dump pocket availability directly propagates to upstream hauling and downstream processing.

A persistent operational challenge is ore accumulation, bridging, and blockage in and around dump pockets, which can force partial or full stoppages to restore material flow. In large-scale operations, blocked-crusher events cause 50–200 hours of unplanned downtime annually, with individual clearing procedures requiring 2–6 hours per incident; at primary gyratory throughputs of 2,000–6,000 t/hr [6], this corresponds to approximately 100,000–1,200,000 tonnes of unprocessed ore per year, equivalent to several days of full production foregone. The disruption extends beyond lost crushing capacity: when the primary hopper is unavailable, loaded haul trucks must divert to a run-of-mine stockpile and complete a second full loading-and-hauling cycle before material reaches the crusher. This diversion can also affect ore tracking and blending control, increasing the risk of dilution or grade variability before processing. Since haulage constitutes approximately 40% of mining operating expenditure [16], this double-handling imposes a substantial incremental cost on the affected tonnes. From an operations perspective, industry handbooks highlight that even a one-percentage-point increase in process availability yields a 4.3% improvement in operating profit [6].

Current dump pocket cleaning and blocked-crusher clearing procedures are frequently performed under constrained access and harsh sensing conditions (dust, poor visibility, irregular ore geometry), and are often only partially mechanized (e.g., breaker booms or excavator-assisted probing). However, authoritative safety guidance emphasizes that blockage clearance should be performed from a position of safety and should *not* involve anyone entering or being lowered into the crushing area due to the potential for sudden, uncontrolled release of stored energy and falling material [2]. Consistent guidance for crushing operations also stresses eliminating routine presence on crusher access platforms and emphasizes prevention and safer clearing practices [3]. Despite these directives, fatal accidents continue to occur when personnel enter hoppers or confined spaces to clear obstructions; Mine Safety and Health Administration (MSHA) reports include cases where a worker entered a hopper to clear blocked material and was engulfed by dumped material [4], and MSHA safety alerts document multiple fatalities from engulfment while clearing obstructions in hoppers/bins/crushers [5].

While partial solutions exist (e.g., rock-breaker booms, probing, or localized "softening"/clearing strategies), they do not provide an integrated pipeline for *autonomous* detection and removal of heterogeneous ore build-up under mine-realistic variability.

Related work. Kim & Choi [15] demonstrated laboratory-scale hopper unblocking using RGB-D vision and a modular detect-then-clear pipeline. ExACT [8] applied Action Chunking with Transformers to control a full-scale excavator with LiDAR+camera fusion for general digging tasks. Our work differs by: (i) benchmarking four IL architectures under identical conditions rather than a single method, (ii) targeting dump pocket cleaning, a safety-critical confined-space task with documented fatalities, and (iii) providing a low-cost ($250), reproducible testbed enabling systematic IL research in mining contexts where full-scale excavators are cost-prohibitive for academic experimentation.

2. OBJECTIVES AND SCOPE

The primary objective is to develop and benchmark an autonomous excavator system in a controlled dump-pocket testbed. This research addresses the "Technology" and "Transformation" themes of the conference by: (i) establishing a reproducible benchmark to quantify removal rate and autonomy reliability, and (ii) supporting progress toward reduced human exposure in hazardous confined spaces. Target KPIs are: *removal rate* (material removed in g/min; maximize autonomous throughput), *success rate* (autonomous session completion without intervention; reliability for deployment), and *safety proxy* (human exposure time in hazard zone; target zero).

3. METHODOLOGY OR APPROACH

The system (Figure 1) is built around two SO-ARM100 robot arms [10] in a leader-follower teleoperation configuration. The SO-ARM100 is a low-cost (sub-$250), open-source, 6-DoF robot arm with 3D-printed PLA components and off-the-shelf STS3215 servo motors (7.4V), developed as an accessible platform for robotics research. The leader arm enables intuitive human demonstration via backdriving: the operator physically moves the leader, and joint angles map to the follower in real-time at 30 Hz. The follower arm executes the task equipped with a custom excavator bucket gripper. The teleoperation interface displays dual-camera streams (wrist and overhead) alongside real-time motor data to enable precise demonstration collection. This setup produced the 162-episode visuomotor dataset used for all experiments. Policy training leverages an open-source imitation learning library [9] with unified infrastructure across all evaluated architectures.

3.1 Task definition

Workspace: Dump pocket testbed with internal box dimensions $L \times W \times H = 0.454 \times 0.310 \times 0.112\,m$. Walls are vertical with inclination $\theta = 90^\circ$ and maximum wall height $H_{\text{max}} = 0.112\,m$.

Target region: The target region is not pre-defined as a fixed ROI; instead, the learned policy learns the cleaning objective implicitly from teleoperation demonstrations.

Material: Natural bentonite clay (commercial cat litter, 0.5–2.5 mm particle size). The material exhibits granular behavior with slight interparticle cohesion, providing stable scooping dynamics. Bulk density $\rho_{\text{bulk}} \approx 1500\text{–}1700\,kg/m^3$; testing conducted under dry indoor conditions.

Initial conditions: Each run begins with 3000 g of material in the dump pocket (± 20 g variation, $<1\%$). The excavator bucket has 18 g max capacity per scoop. The evaluation measures continuous removal rate (g/min); complete pocket evacuation within the session is not required.

Task cycle: The policy detects regions requiring cleaning, approaches with the bucket, scoops material (18 g max capacity), transports to the primary crusher at the center of the dump pocket, and deposits the material. This cycle repeats continuously to maximize removal rate.

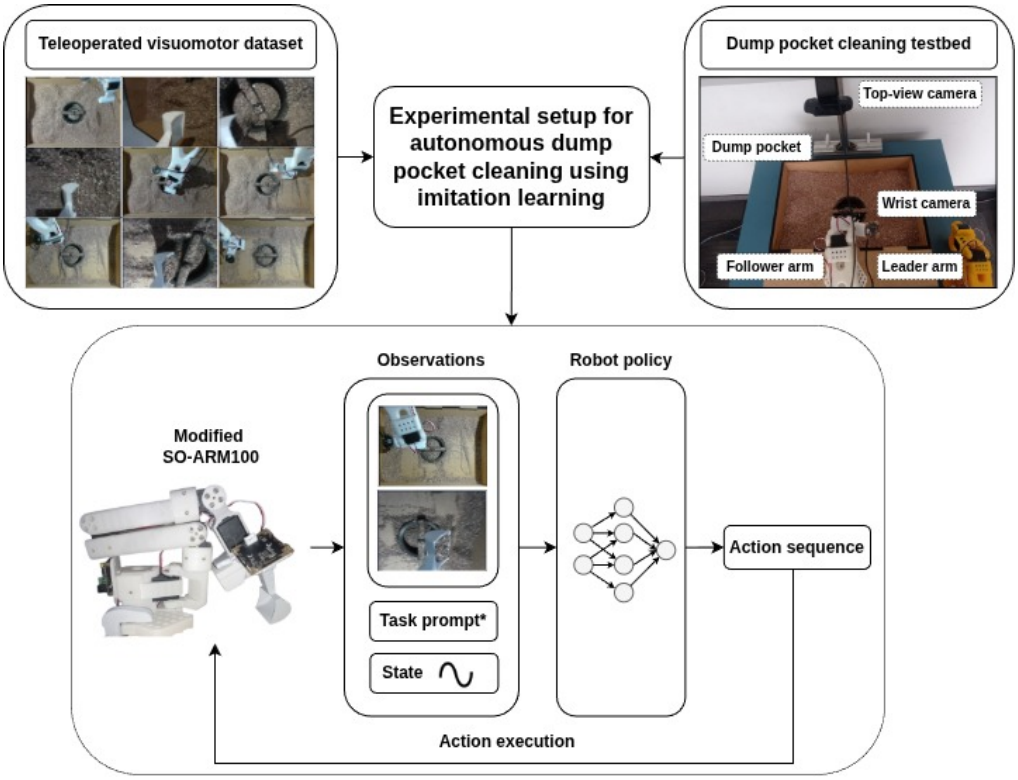

Figure 1 – Experimental setup for autonomous dump pocket cleaning using imitation learning. Top left: teleoperated dataset (162 demonstrations). Top right: physical testbed with dual RGB cameras (top-view, wrist-mounted), dump pocket, and SO-ARM100 platform (follower and leader arms). Bottom: imitation learning pipeline receiving observations (camera streams, task prompt*, proprioceptive state), processing through policy network, and executing actions in closed loop. *Task prompt: "use the scoop-like end effector to gather material from the dump pocket and deposit it into the primary crusher at the center", used only by VLA models ($\pi_{0.5}$, SmolVLA) for language conditioning; ACT and Diffusion operate on vision and proprioception only.

Success criteria: We define *success* operationally as (i) fully autonomous operation without human intervention during the 10-minute session, (ii) no safety-limit violations, and (iii) measurable material removal. All methods start from the same initial state (3 kg material mass), and the primary performance metric is total *material removed* (g/10 min).

3.2 Observations and Action Space

We define the observation o_t as the concatenation of two RGB streams and proprioception:

- **Wrist camera:** 2MP camera (30 FPS) mounted on the follower arm end-effector (bucket), providing close-range perspective of material and manipulation area.
- **Overhead camera:** Webcam providing global context view of the dump pocket and arm workspace.

- **Proprioception:** Joint angles (6-DoF from STS3215 servos) and gripper state, synchronized with camera timestamps at 30 Hz.
- **Task prompt (VLA only):** Natural language instruction: "use the scoop-like end effector to gather material from the dump pocket and deposit it into the primary crusher at the center", used by $\pi_{0.5}$ and SmolVLA for multimodal conditioning; ACT and Diffusion do not use language input.

The action a_t is a vector of joint-position targets q_{target} predicted by the policy and executed by the low-level servo controller. Actions are produced at 30 Hz control frequency. ACT uses action chunking with horizon $H = 100$ steps; $\pi_{0.5}$, SmolVLA, and Diffusion Policy operate in receding-horizon mode with model-specific temporal windows as configured in the training library [9]. Table 1 summarizes the platform specifications and dataset statistics.

Table 1 – Platform and dataset summary.

Platform	Dataset
SO-ARM100, 6-DoF, bucket 0–18 g	162 eps (129/33 split), 73,944 frames
Sub-$250, leader-follower	30 FPS, mean 15.18 s/ep (range 10.4–26.6 s)
Dual RGB (wrist + overhead)	40.99 min total, expert rate 62.04 g/min

3.3 Models

The control is governed by four imitation learning paradigms: **ACT** (Action Chunking with Transformers) uses a CVAE to predict action chunks trained from scratch with an $L_1 + \lambda KL$ objective [7, 8]; $\pi_{0.5}$ ($\sim$3.7B-parameter VLA) employs hierarchical inference, with high-level subtask prediction followed by flow-matching action generation, pretrained on heterogeneous multi-source data (household manipulation, cross-embodiment, web data) [12, 11]; **SmolVLA** (450M-parameter compact VLA) uses a flow-matching Action Expert pretrained on 481 open-source community manipulation datasets [13]; and **Diffusion Policy** formulates action generation as conditional denoising diffusion via an MSE noise-prediction objective with iterative Stochastic Langevin Dynamics at inference [14].

3.4 Experimental Protocol

Baseline (Expert Teleoperation): Expert human teleoperation serves as the gold standard. Across 162 demonstration episodes (40.99 min total), the expert achieved 62.04 g/min, yielding an estimated 620 g over a 10-minute session, the performance upper bound under current hardware constraints.

Evaluation protocol: Each policy was evaluated in a single 10-minute continuous autonomous session (3 kg material mass, fixed lighting). The primary metric is total material removed, measured by weighing the outlet collection tray before and after each session. Given N=1 session per model, we report observed total removed mass and equivalent removal rate (g/min) without statistical aggregation.

Offline training: All models were trained on the 129-episode dataset using an open-source IL library [9]. Best checkpoints selected by validation loss: ACT at step 9205 (val loss 0.2746), $\pi_{0.5}$ at step 25774 (0.0745), SmolVLA at step 14728 (0.0254), Diffusion at step 7252 (0.0141). VLA models ($\pi_{0.5}$, SmolVLA) leverage pretrained backbones [12, 13]; ACT and Diffusion trained from scratch. Since ACT optimizes a composite objective ($L_1 + \lambda KL$) while other policies use MSE/flow-matching losses, Figure 2(a) plots ACT as $(\text{val/loss})^2$ for visual scale comparability (monotone transformation; checkpoint ranking preserved); this panel is a convergence diagnostic, not a cross-architecture ranking.

4. RESULTS, OUTCOMES, OR PERFORMANCE

Results are summarized in Figure 2 and Table 2. All models were evaluated in a single 10-minute session under identical initial conditions (3 kg bentonite, fixed lighting, indoor laboratory setting).

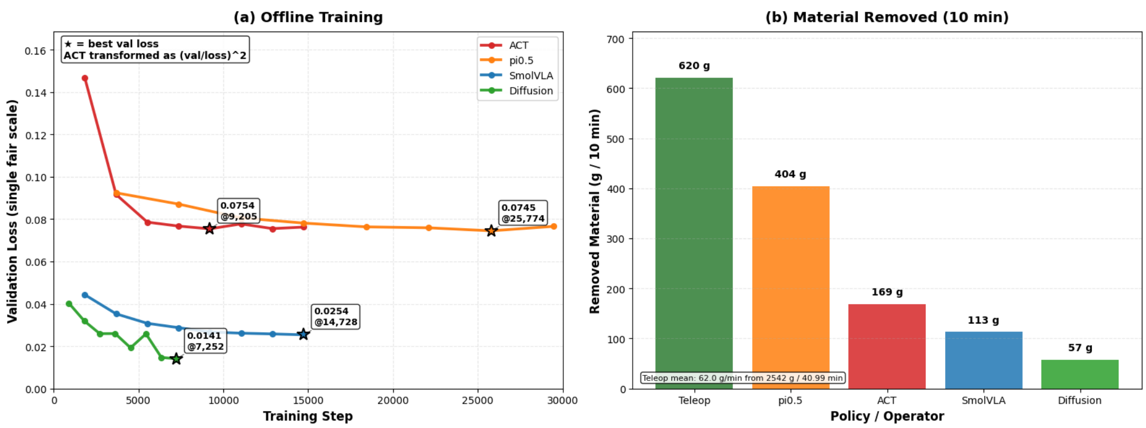

Figure 2 – Benchmark results and offline training metrics. (a) Validation loss trajectories; ACT is plotted as $(\text{val/loss})^2$ for visual scale comparability (see Section 3 for justification). Best checkpoints (stars): ACT 0.0754 @ 9,205 steps; $\pi_{0.5}$ 0.0745 @ 25,774 steps; SmolVLA 0.0254 @ 14,728 steps; Diffusion 0.0141 @ 7,252 steps. This panel is a convergence diagnostic; cross-architecture comparison uses real-robot results only. (b) Total material removed (g) in a single 10-minute autonomous session. Expert baseline ($\approx$620 g) estimated from demonstration data (62.0 g/min, 2542 g over 40.99 min). $\pi_{0.5}$ removes 404 g, followed by ACT (169 g), SmolVLA (113 g), and Diffusion Policy (57 g).

Note on statistical power: This evaluation reports single observed values per model (N=1 session); statistical aggregation is not applicable. Results represent exploratory observations pending systematic replication.

Failure observations: Qualitative rollout observations included partial scoops, wall adhesion, bucket slip, and occasional hesitation; these were not quantified in this preliminary evaluation.

Key findings: Under controlled laboratory conditions, $\pi_{0.5}$ achieves the highest removal performance (404 g in 10 minutes, 40.4 g/min), reaching approximately 65% of the expert teleoperation estimate (620 g). ACT (169 g, 27% of expert) ranks second despite having no pretrained representations, trained solely on 162 demonstration episodes.

Table 2 – Benchmark results: total material removed in a single 10-minute autonomous session. Expert baseline estimated from demonstration data (62.04 g/min $\times$ 10 min). All learned policies evaluated under identical initial conditions (3 kg bentonite, fixed lighting). Given N=1 session per model, statistical aggregation (mean$\pm$std) is not applicable; values represent single observed measurements.

Model	**Removed (g / 10 min)**	**Rate (g/min)**	**% of Expert (est.)**
Expert Teleoperation (est.)	$\approx$620	62.0	100
$\pi_{0.5}$	404	40.4	65
ACT	169	16.9	27
SmolVLA	113	11.3	18
Diffusion Policy	57	5.7	9

SmolVLA (113 g, 18% of expert) ranks third, removing less material than ACT despite being pretrained on 481 open-source community manipulation datasets. This single-session result raises the hypothesis that for narrow single-task settings where the target domain differs from pretraining data, learning from scratch may remain competitive with fine-tuning a pretrained model. Diffusion Policy (57 g, 9% of expert) removes the least material, consistent with the slower reactive behavior observed during rollout.

5. DISCUSSION

The benchmark demonstrates laboratory feasibility of learned policies for autonomous material removal in a simplified dump-pocket analogue. A key interpretive hypothesis is that pretraining benefit may depend on domain proximity and model capacity: $\pi_{0.5}$ (3.7B parameters, broadly pretrained across diverse robot embodiments) achieves the highest observed removal rate in this novel setting, while ACT trained from scratch outperforms SmolVLA (450M parameters, pretrained narrowly on household manipulation) in the single-session benchmark. This result suggests a hypothesis for future replicated testing: in narrow single-task settings where the target domain differs from pretraining data, learning from scratch may remain competitive with fine-tuning a pretrained model. Results reflect simplified laboratory conditions (bentonite, fixed lighting, N=1 session per model) that may not stress-test reliability differences emerging under operational challenges.

Model performance analysis: $\pi_{0.5}$'s leading observed performance may be related to several factors: (i) *scale*: $\sim$3.7B parameters trained for 25,774 steps, the most compute-intensive configuration evaluated; (ii) *hierarchical reasoning*: it decouples high-level subtask prediction from low-level action generation, which may support adaptation to varying material states; and (iii) *pretraining breadth*: its heterogeneous multi-source data (household, cross-embodiment, web) may provide richer priors than SmolVLA's 481 community manipulation datasets, neither of which includes mining excavation data. These factors are consistent with the observed ranking, but isolating their individual effects requires replicated testing and targeted ablations.

Deployment barriers: (i) *Material variability*: bentonite testbed versus heterogeneous ore (dust to boulders); (ii) *Scale*: testbed 1000× smaller than industrial hoppers (0.016 vs 10–50 m^3); (iii) *Environment*: dust, poor lighting, vibration will degrade vision [17]; (iv) *Control responsiveness*: Diffusion Policy exhibited visibly slower reactive behavior compared to other methods, consistent with the iterative denoising inference process requiring multiple forward passes per action; and (v) *Data acquisition*: scaling to industrial embodiments requires new demonstrations on the target hardware; the algorithmic workflow may transfer, but final deployment data must be collected at the relevant scale and hardware. This lab-to-mine gap is the primary limitation.

Task-state ambiguity and visual activation: ACT, SmolVLA, and Diffusion Policy exhibited hesitation at rollout start: the initial observation closely resembles the terminal state (same rest pose, similar material distribution), creating visual ambiguity between "task just started" and "task finished." $\pi_{0.5}$ did not exhibit this, likely due to language conditioning. A practical mitigation is a visual activation cue (e.g., a color indicator at rollout start) to help vision-only policies distinguish task states without architectural changes.

Dataset scale: The 129-episode training split is modest by IL standards; scaling to several hundred demonstrations with greater material-state variability would likely improve performance across all architectures and remains a prerequisite before drawing firm conclusions about their relative upper bounds.

6. CONCLUSIONS AND IMPLICATIONS FOR INDUSTRY

This work introduces an experimental testbed and open benchmark for evaluating imitation learning architectures on mining excavation tasks under controlled laboratory conditions. In a single preliminary 10-minute evaluation session, $\pi_{0.5}$ removes 404 g (65% of the expert teleoperation estimate) while ACT, SmolVLA, and Diffusion Policy remove 169, 113, and 57 g respectively. A notable observation is that ACT, trained from scratch on 162 demonstrations, outperforms SmolVLA despite its pretraining on community manipulation datasets in this single-session benchmark. This raises the hypothesis that for narrow single-task settings where the target domain differs from pretraining data, learning from scratch may remain competitive with fine-tuning a pretrained model. These preliminary results establish an experimental foundation and open dataset to support future research on autonomous tasks for the mining industry.

The primary practical implication is safety: these results demonstrate laboratory feasibility of autonomous material-removal behavior in a simplified dump-pocket analogue, addressing the engulfment risk that manual clearing poses to workers in confined hoppers and crusher areas [4, 5, 2]. Substantial barriers remain before industrial deployment: the testbed uses bentonite rather than heterogeneous ore, operates at 1000× smaller scale (0.016 vs. 10–50 m^3), lacks dust and vibration, and would require new demonstrations on the target industrial embodiment; the algorithmic workflow may transfer, but final deployment data must be collected at the relevant scale and hardware. This lab-to-mine gap is the primary limitation. Given the downtime and haulage costs documented in Section 1, partial automation could be economically meaningful, but quantifying return on investment requires mine-scale validation.

From a research perspective, this work establishes a precedent for introducing IL methods to mining contexts, with applicability beyond dump pocket cleaning: the IL framework is task-agnostic, and manipulation tasks demonstrable via teleoperation may be studied using the same pipeline, suggesting relevance to ore pass clearing, secondary crusher feeding, and similar confined-space operations. This benchmark provides: (i) an accessible lab-scale dump-pocket testbed built on the open-source SO-ARM100 low-cost robot platform ($250 hardware), (ii) an open dataset (162 episodes) for reproducible comparisons, and (iii) a preliminary hypothesis that pretraining benefit depends on domain proximity. Concrete next steps include scaling the demonstration dataset with greater material variability, validating with industrial-scale arms, and testing under dust and vibration conditions.

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